

Product Owner Candidate Exercises

Table of Contents

General information	1
Exercise instructions	2
Fault Assessment Calculator	3
Questions	4
Vehicle Lookup	5
Epic	5
Summary	5
Use Case	5
Acceptance Criteria	6
Business Requirements	6
Background	6
Key Search Methods	6
Vehicle Lookup by Vehicle Model Code	7
Vehicle Lookup by Make and Model	8
Other considerations	9
Questions	11

General information

One of the key responsibilities of the Product Owner is acting as a bridge between product management and engineering. This responsibility requires the Product Owner to work closely with the business analyst and technical leads to effectively capture the business intent in business stories in a way that is understood by the engineering teams that will implement them.

The aim of these exercises is to role play a business story review. In attendance there will be:

- The product manager
- The business analyst
- The technical lead and/or architect

User story writing exercise

The exercises provide background and epic/feature descriptions with additional business and other information required to write stories for the development team. From these stories the development team will create implementation tasks.

The general process used in Global Content at Guidewire for getting to story creation is:

- The product manager identifies the broad market requirement and breaks down that market requirement into features to realise that requirement
- The business analyst under the product managers direction investigates areas that require additional detail for implementation and captures the information in requirements documents
- With the Business Analyst, the Product Owner identifies the scenarios that become the basis of the business stories
- The Product Owner creates first drafts of the business stories
- The Product Owner finalises the business story in collaboration with the technical lead and business analyst.

Exercise instructions

Write the first draft user stories for the Epics aligning to the template provided. The user story should include the following information:

- User role (Persona): The role of the target user (user, administrator, etc.)
- Purpose: The function or action the user wants to achieve
- Value gained: The value that the function or action brings to the user
- Acceptance criteria (definition of Done): The exit conditions for this story
- Add other information to the story as needed. This could include elements such as screens, flows, data elements etc.
- Also include notes that would be helpful when finalising with the lead engineer

Please explain if there is additional information that you need to complete the stories.

Suggest a priority for the stories using MoSCoW and explain your reasoning for your priorities.

Fault Assessment Calculator

Read the following Japanese and the attached PDF (HanreiTimesSamplePages.pdf) and answer questions.

Regarding this exercise, please write all presentation materials, such as epics and user stories, in Japanese.

Also, please make your presentation in Japanese.

Business Issues

交通事故の過失割合は、事故状況から割り出した認定基準をもとに、修正要素を考慮して決定している。具体的には、別冊判例タイムズに掲載された認定基準を参考にしている。そして、その別冊判例タイムズには、様々な事故形態・状況における認定基準が300件以上掲載されている。

<https://www.hanta.co.jp/books/6137/>

例えば、交差点での歩行者と直進車との事故であれば、添付PDF (HanreiTimesSamplePages.pdf) の67ページを参考にする。

まず、基本過失割合を決定する。この事例の場合は10%になる。その値に対して、修正要素を考慮する。例えば、歩行者が幼児の場合、上の基本過失割合に対して「-5%」の調整を行い、最終の過失割合は5%となる。

現在、このプロセスは手作業で行うため、認定基準を探し出すために多くの手間がかかっている。また、人によって選択する認定基準や割り出した過失割合の値にバラツキが見られる。そこで、物理的な本を閲覧することなく、オンラインで認定基準を検索して過失割合を決定する機能を追加したい。

なお、数百の認定基準データの中から、該当の事故状況にマッチした認定基準を効率的かつ的確に検索する工夫も考慮すること。(具体的な認定基準の分類やタイトルは添付PDFの25から42ページにある第1章から第7章までの目次を参照)

As a result of analyzing the above business issues, we decided to develop the following epics:

Epic 1: 認定基準の検索と過失割合の算出

Goal: 別冊判例タイムズの事故類型分類に基づいた認定基準を構造化し、効率的な検索をオンライン上で可能にする。

Business background: 現在、この作業は物理的な本を使用して行われており、検索に時間がかかる。

Epic 2: AI支援による事故状況の自動分類

Goal: 事故報告の内容から、AIが自動的に最も適合する認定基準を提案する。

Business background: 利用者の経験により、算出された過失割合の値にバラツキが見られる。

Questions

1. Create user stories that realize the features defined in Epics 1 and 2. Also, if there are any other features you think will be necessary in the first release, create epics and user stories for them.
2. Prioritize the user stories and choose the minimum stories needed for your first release. (MVP)

Vehicle Lookup

Read the following Epic description and answer the questions that follow.

Epic

Summary

The feature allows a policy buyer to provide a unique vehicle model code to the underwriter. Each model includes its own rating factors; therefore, it is important for underwriters to know the exact model code.

This feature will make use of a sample data source for the purposes of testing and demonstration. An implementer is expected to replace the sample data source with an actual data source that conforms to the data contract and API.

Use Case (End user = Policy Underwriter)

1. Look up the vehicle master data by Vehicle Model Code (型式). This code is written on the Vehicle Inspection Certificate, so I want to ask the policy buyer if he knows this code.
 - a. If the answer is Yes, then I use the code to look up the particular model from the vehicle master. If there's no match.... go to (2)
 - b. If the answer is No, then go to (2)
2. Look up the vehicle master data by Vehicle Make & Model name. This is probably the most likely path. I want to ask the policy buyer...
 - a. First Registration Year & Month (初度登録年月; user to filter the model name)
 - b. Make of the vehicle (メーカー; e.g. Toyota, Honda, etc.)
 - c. Model name of the vehicle (車名; e.g. 2020 Accord, 2021 Civic). The list of model names should be filtered by the first registration date and by the choice of the Make
 - d. I must choose the exact model name of the insured vehicle from the search result.
3. As a result of the search, either from (1) or (2), I want the following info to be copied into the insured vehicle data on PolicyCenter. Note that the master data gets updated regularly, and I want the vehicle information as of the policy start date.
4. As an underwriter, after the attempt of (1) and (2), if the vehicle model code of the insured vehicle is not found in the vehicle master data, I want to create a new model in order to proceed with the quote. This scenario might happen if the insured vehicle is a foreign car, or it has been just released recently and not on the vehicle master yet. I want to capture the following information for the non-existing model:

- a. Make (e.g. "Ford")
- b. Model Name (e.g. "マダナイ")
- c. Vehicle Model Code (e.g. "UNKNOWN")
- d. New Cost (e.g. JPY 2,000,000)
- e. Price Range Max (e.g. JPY 2,000,000)
- f. Price Range Min (e.g. JPY 1,500,000)
- g. Injury Liability Class (e.g. 1)
- h. Property Liability Class (e.g. 2)
- i. Passenger Injury Class (e.g. 3)
- j. Vehicle Damage Class (e.g. 9)
- k. Vehicle Use Type(a typekey from all the vehicle user types)
- l. Release Date (a date when this car has become available)

Acceptance Criteria

- UI for Underwriter to choose the exact vehicle model:
 - Enter First Registration Date (just Year and Month, no Day)
 - Enter vehicle model code if it is known
 - Otherwise, search by Make and Model name
 - Select the exact Model from the search result
- UI for Underwriter to add a new model in an ad hoc way
 - All vehicle details above can be entered for a new model
- When underwriter selects a particular vehicle model (or create a new model), the vehicle info is copied from the vehicle master to the policy's vehicle data

Business Requirements

Background

The vehicle lookup system is designed to facilitate efficient vehicle information retrieval during insurance policy submission through two primary search methods and several supporting features:

Key Search Methods

Vehicle Model Code Search

- Allows precise vehicle identification using unique model codes

- Provides exact matches from the Cognivision vehicle price master database
- Returns comprehensive vehicle specifications and pricing information

Make and Model Search

- Alternative method when model codes are unknown
- Enables broader searches using manufacturer and model names
- Returns lists of matching vehicles with detailed specifications

Core Features:

- Dynamic data filtering based on vehicle registration dates
- Intelligent dropdown menus to prevent information overload
- Type-ahead search functionality for improved user experience
- Support for both English and Japanese character searches
- Automatic mapping of vehicle information to insurance policies

Data Management:

- Integration with Cognivision vehicle price master database
- Capability to create and store new vehicle models
- Repository for non-standard vehicle specifications
- Regular synchronization of vehicle master data

Vehicle Lookup by Vehicle Model Code

Search by vehicle model code:

- If exact / partial match, then return vehicle info of matching vehicle model(s)
- If no match, return nothing (+ display message "No results found")

Retrieve information about the vehicle from the vehicle master (to begin with, sample data) and map this information to the insured vehicle

- Make (メーカー);
- Model (モデル);
- Vehicle Model Code (型式コード);
- New Vehicle Cost (新車価格);

- Price Range Max (最高車両価格);
- Price Range Min (最低車両価格);
- Bodily Injury Liability Class (自動車対人賠償クラス);
- Property Damage Liability Class (自動車対物賠償クラス);
- PIP / Passenger Injury Liability Class (自動車対物賠償クラス);
- Vehicle Damage Class (自動車車両損害クラス);
- Vehicle Use Type (用途車種);
- Release Date (発売日)

This search method enables underwriters to locate vehicle information during policy submission using a specific vehicle model code.

Each **model code** uniquely identifies vehicles with identical specifications, structure, equipment, and performance.

When searching with a model code, the system queries the Cognivision vehicle price master database to find an exact match.

The system will display the matching vehicle's detailed information as of the policy start date.

If no match is found, a "No results found" message appears on the screen.

Once a vehicle is selected, the system automatically maps all relevant vehicle information to the policy. In cases where the policy start date falls between two master data update dates, the system uses the most recent value to ensure accuracy in policy rating and claims processing.

Vehicle Lookup by Make and Model

Search by make and model:

- Return a list of vehicles matching the make and model in search criteria
- If no matches, return nothing (+ display message "No results found")

Retrieve information about the vehicle from the vehicle master (to begin with, sample data) and map this information to the insured vehicle

- Make (メーカー);
- Model (モデル);

- Vehicle Model Code (型式コード);
- New Vehicle Cost (新車価格);
- Price Range Max (最高車両価格);
- Price Range Min (最低車両価格);
- Bodily Injury Liability Class (自動車対人賠償クラス);
- Property Damage Liability Class (自動車対物賠償クラス);
- PIP / Passenger Injury Liability Class (自動車対物賠償クラス);
- Vehicle Damage Class (自動車車両損害クラス);
- Vehicle Use Type (用途車種);
- Release Date (発売日)

This alternative search method is particularly useful when the vehicle model code is unknown.

Underwriters can search the database using a combination of the **manufacturer name (make) and model name**, such as "Toyota Civic."

The system queries the Cognivision vehicle price master database and returns a list of all possible matching vehicles.

Users can view the detailed information for each matching vehicle as of the policy start date. If no matches are found, the system displays a "No results found" message.

From the list of results, underwriters can select the appropriate vehicle, and the system will map all relevant information to the policy.

As with the model code search, if the policy start date falls between two master data update dates, the system uses the most recent value to maintain accuracy in policy documentation.

Other considerations

If we simply import the list of vehicles from the master data and use it to feed the dropdowns for Make and Model, the list could run several pages long. This is inefficient, and is certainly not user-friendly system behaviour. Therefore, we must find a way to limit the entries that appear in the dropdowns.

The master data contains information such as release (sale) date, make, model, etc. of each vehicle model.

The policyholder, of course, knows the first registration date of their vehicle (in their vehicle inspection certificate) and the names of the make and model of their vehicle.

We use this information to filter out irrelevant vehicle models to make the search process smooth and user-friendly.

The requirement is to filter out any vehicle models released (i.e., came into the market for the first time) after the first registration date of the insured vehicle, and to filter out model names not belonging to the selected vehicle make.

Specifically:

- If first registration date of a vehicle is date D; then update the vehicle make and model dropdowns to only display those released on or before date D
- If a vehicle make is specified (say, Toyota), then update the vehicle model name dropdown to only display vehicle models belonging to make Toyota.

In some cases, while the vehicle itself exists, its detailed specifications, as identified by the model code, might not be standardized in the typical "vehicle master data" used by insurers.

In such cases, the user must have the option of creating a new vehicle model. The user can create the model on the Create Vehicle Model UI by entering the information related to the fields mentioned in summary.

It is possible that this information could be retained for future use. Insurers using PolicyCenter may also decide to regularly sync the created vehicle models with up-to-date information in the vehicle master. Alternatively, they may retain this information in order to update their vehicle master data (either manually or by configuring some logic).

If an insured vehicle is already selected, confirm whether the user wants to overwrite this before proceeding to create the new vehicle model.

The requirement is:

- Develop UI for Create New Vehicle Model
- Map this information to the insured vehicle
- Store this information in the policy file
- Store this vehicle model in a repository for future use (either to sync or update vehicle master data)

The number of distinct possible values in the field make (メーカー), and especially in model (モデル) and vehicle model code (型式コード) is very high. In order to improve the search experience, we shall implement type-ahead search functionality in these fields.

The requirement is

- When the user is performing a search for a particular field, he/she types a series of characters which match entries in the master data
- The list / dropdown should automatically display the likely entries based on the search string
- The list should be sorted in the following order (top to bottom):
 - Entries starting with the search string
 - Entries having the search string somewhere in the middle
 - Entries ending with the search string

This behaviour may not be possible in case of Japanese search.

Japanese search applies to make (メーカー) and model (モデル), which can be searched using Japanese characters (eg. トヨタ for Toyota; ホンダ for Honda). The user may choose to enter a partial search string (eg. トヨ “toyo” for トヨタ; プリ “puri” for プリウス) or the full search string.

Vehicle Model Code (型式コード) will always be a set of alphanumeric characters, so isn't a target for Japanese search.

In case of Japanese search, the following is the expected behaviour:

- User enters the search string in Japanese (full text search or partial match search)
- User clicks "Search"
- A list of possible matches appears, sorted in Japanese alphabetical order

Questions

1. Write user stories to break down the Epic, keeping in mind the overall Business Requirements.
2. Prioritize the user stories and choose the minimum stories needed for your first release. (MVP)
3. What are the key considerations from an implementation perspective?
4. How would you address any challenges you foresee from an implementation perspective? Pick any one key consideration from answer 3 and discuss the potential challenges you foresee.

Structure your answers bearing in mind Agile development principles and regular opportunities to interact with end-users.